# Setting up Prometheus — NVIDIA GPU Telemetry 1.0.0 documentation

17–22 minutes

---


- About Setting up Prometheus
- Setting up DCGM
- DCGM Exporter Helm Chart Customization
- Using Grafana
- Patching the Grafana Service
- Port Forwarding
- DCGM Dashboard in Grafana
- Viewing Metrics for Running Applications


## About Setting up Prometheus

Implementing a Prometheus stack can be complicated but can be managed by taking advantage of the `Helm` package manager and the Prometheus Operator and kube-prometheus projects. The

Operator uses standard configurations and dashboards for Prometheus and Grafana and the Helm prometheus-operator chart allows you to get a full cluster monitoring solution up and running by installing Prometheus Operator and the rest of the components listed above.

First, add the `helm` repo:

```
$ helm repo add prometheus-community \
  https://prometheus-community.github.io/helm-charts
```

Now, search for the available `prometheus` charts:

```
$ helm search repo kube-prometheus
```

After you locate the version of the chart to use, inspect the chart so we can modify the settings:

```
$ helm inspect values prometheus-community/kube-prometheus-stack > /tmp/kube-prometheus-stack.values
```

Next, edit the values file to change the port at which the Prometheus server service is available. In the `prometheus` instance section of the chart, change the service type from `ClusterIP` to `NodePort`. This will allow the Prometheus server to be accessible at your machine ip address at port 30090 as `http://<machine-ip>:30090/`

From:

```
## Port to expose on each node
## Only used if service.type is 'NodePort'
##
nodePort: 30090

## Loadbalancer IP
```

```
## Only use if service.type is "loadbalancer"
loadBalancerIP: ""
loadBalancerSourceRanges: []
## Service type
##
type: ClusterIP
```

To:

```
## Port to expose on each node
## Only used if service.type is 'NodePort'
##
nodePort: 30090

## Loadbalancer IP
## Only use if service.type is "loadbalancer"
loadBalancerIP: ""
loadBalancerSourceRanges: []
## Service type
##
type: NodePort
```

Also, modify the `prometheusSpec.serviceMonitorSelectorNilUsesHelmValues` settings to `false` below:

```
## If true, a nil or {} value for prometheus.prometheusSpec.serviceMonitorSelector will cause the
## prometheus resource to be created with selectors based on values in the helm deployment,
```

```
## which will also match the servicemonitors created
##
serviceMonitorSelectorNilUsesHelmValues: false
```

Add the following `configMap` to the section on `additionalScrapeConfigs` in the Helm chart.

```
## AdditionalScrapeConfigs allows specifying additional Prometheus scrape configurations. Scrape configurations
## are appended to the configurations generated by the Prometheus Operator. Job configurations must have the form
## as specified in the official Prometheus documentation:
## https://prometheus.io/docs/prometheus/latest/configuration/configuration/#scrape_config. As scrape configs are
## appended, the user is responsible to make sure it is valid. Note that using this feature may expose the possibility
## to break upgrades of Prometheus. It is advised to review Prometheus release notes to ensure that no incompatible
## scrape configs are going to break Prometheus after the upgrade.
##
## The scrape configuration example below will find master nodes, provided they have the name .*mst.*, relabel the
## port to 2379 and allow etcd scraping provided it is running on all Kubernetes master nodes
##
additionalScrapeConfigs:
- job_name: gpu-metrics
  scrape_interval: 1s
```

```
  metrics_path: /metrics
  scheme: http
  kubernetes_sd_configs:
  - role: endpoints
    namespaces:
      names:
      - gpu-operator
  relabel_configs:
  - source_labels: [__meta_kubernetes_endpoints_name]
    action: drop
    regex: .*-node-feature-discovery-master
  - source_labels: [__meta_kubernetes_pod_node_name]
    action: replace
    target_label: kubernetes_node
```

Finally, deploy the Prometheus and Grafana pods using the `kube-prometheus-stack` via Helm:

```
$ helm install prometheus-community/kube-prometheus-stack \
  --create-namespace --namespace prometheus \
  --generate-name \
  --values /tmp/kube-prometheus-stack.values
```

Note

You can also override values in the Prometheus chart directly on the Helm command line:

```
$ helm install prometheus-community/kube-prometheus-stack \
  --create-namespace --namespace prometheus \
```

```
  --generate-name \
  --set prometheus.service.type=NodePort \
  --set prometheus.prometheusSpec.serviceMonitorSelectorNilUsesHelmValues=false
```

You should see a console output as below:

```
NAME: kube-prometheus-stack-1637791640
LAST DEPLOYED: Wed Nov 24 22:07:22 2021
NAMESPACE: prometheus
STATUS: deployed
REVISION: 1
NOTES:
kube-prometheus-stack has been installed. Check its status by running:
  kubectl --namespace prometheus get pods -l "release=kube-prometheus-stack-1637791640"

Visit https://github.com/prometheus-operator/kube-prometheus for instructions on how to create &
configure Alertmanager and Prometheus instances using the Operator.
```

Now you can see the Prometheus and Grafana pods:

```
$ kubectl get pods -A

NAMESPACE    NAME                                          READY   STATUS    RESTARTS   AGE
kube-system   calico-kube-controllers-8f59968d4-g28x8      1/1     Running   1          23m
kube-system   calico-node-zfnfk                            1/1     Running   1          23m
kube-system   coredns-f9fd979d6-p7djj                      1/1     Running   1          23m
kube-system   coredns-f9fd979d6-qhhgq                      1/1     Running   1          23m
kube-system   etcd-ip-172-31-92-253                        1/1     Running   1          23m
```

```
kube-system   kube-apiserver-ip-172-31-92-253                    1/1   Running   2       23m
kube-system   kube-controller-manager-ip-172-31-92-253           1/1   Running   1       23m
kube-system   kube-proxy-mh528                                   1/1   Running   1       23m
kube-system   kube-scheduler-ip-172-31-92-253                    1/1   Running   1       23m
kube-system   nvidia-device-plugin-1603211071-7hlk6              1/1   Running   0       15m
prometheus    alertmanager-kube-prometheus-stack-1603-alertmanager-0   2/2   Running   0
13m
prometheus    kube-prometheus-stack-1603-operator-6b95bcdc79-wmbkn     2/2   Running   0
13m
prometheus    kube-prometheus-stack-1603211794-grafana-67ff56c449-tlmxc   2/2   Running   0
13m
prometheus    kube-prometheus-stack-1603211794-kube-state-metrics-877df67c49f   1/1   Running
0       13m
prometheus    kube-prometheus-stack-1603211794-prometheus-node-exporter-b5fl9   1/1   Running
0       13m
prometheus    prometheus-kube-prometheus-stack-1603-prometheus-0       3/3   Running   1
13m
```

## Setting up DCGM

Now, we will deploy DCGM Exporter to gather GPU telemetry. First, lets setup the Helm repo:

```
$ helm repo add gpu-helm-charts \
  https://nvidia.github.io/dcgm-exporter/helm-charts
```

And then update the Helm repo:

```
$ helm repo update
```

## DCGM Exporter Helm Chart Customization

The DCGM-Exporter helm package includes several customization options for various use cases.

arguments

Customize the command-line parameters passed to dcgm-exporter on startup.

*Example: Set the metric collection interval to 1000 milliseconds.*

```
arguments["-c", "1000"]
```

extraConfigMapVolumes

Attach ConfigMap volume containing the metrics to the be watched.

*Example: Attach the 'exporter-metrics-config-map' volume to the pod.*

```
extraConfigMapVolumes:
- name: exporter-metrics-volume
  configMap:
    name: exporter-metrics-config-map
```

extraEnv

Customize environment variables, including and especially the DCGM-Exporter variables.

*Example: Collect the metrics specified in the ConfigMap `exporter-metrics-volume`.*

```
extraEnv:
- name: DCGM_EXPORTER_CONFIGMAP_DATA
```

```
        value: "default:exporter-metrics-volume"
```

Install the DCGM Exporter chart:

```
$ helm install \
  --generate-name \
  gpu-helm-charts/dcgm-exporter
```

Now, you can observe the DCGM Exporter pod:

```
$ kubectl get pods -A
NAMESPACE    NAME                                                    READY   STATUS    RESTARTS   AGE
default      dcgm-exporter-2-1603213075-w27mx                        1/1     Running   0          2m18s
kube-system  calico-kube-controllers-8f59968d4-g28x8                 1/1     Running   1          43m
kube-system  calico-node-zfnfk                                       1/1     Running   1          43m
kube-system  coredns-f9fd979d6-p7djj                                 1/1     Running   1          43m
kube-system  coredns-f9fd979d6-qhhgq                                 1/1     Running   1          43m
kube-system  etcd-ip-172-31-92-253                                   1/1     Running   1          43m
kube-system  kube-apiserver-ip-172-31-92-253                         1/1     Running   2          43m
kube-system  kube-controller-manager-ip-172-31-92-253                1/1     Running   1          43m
kube-system  kube-proxy-mh528                                        1/1     Running   1          43m
kube-system  kube-scheduler-ip-172-31-92-253                         1/1     Running   1          43m
kube-system  nvidia-device-plugin-1603211071-7hlk6                   1/1     Running   0          35m
prometheus   alertmanager-kube-prometheus-stack-1603-alertmanager-0  2/2     Running   0
33m
prometheus   kube-prometheus-stack-1603-operator-6b95bcdc79-wmbkn    2/2     Running   0
33m
```

```
prometheus   kube-prometheus-stack-1603211794-grafana-67ff56c449-tlmxc       2/2    Running    0
33m
prometheus   kube-prometheus-stack-1603211794-kube-state-metrics-877df67c49f   1/1    Running
0        33m
prometheus   kube-prometheus-stack-1603211794-prometheus-node-exporter-b5fl9   1/1    Running
0        33m
prometheus   prometheus-kube-prometheus-stack-1603-prometheus-0            3/3    Running    1
33m
```

You can view the services setup as part of the operator and DCGM Exporter:

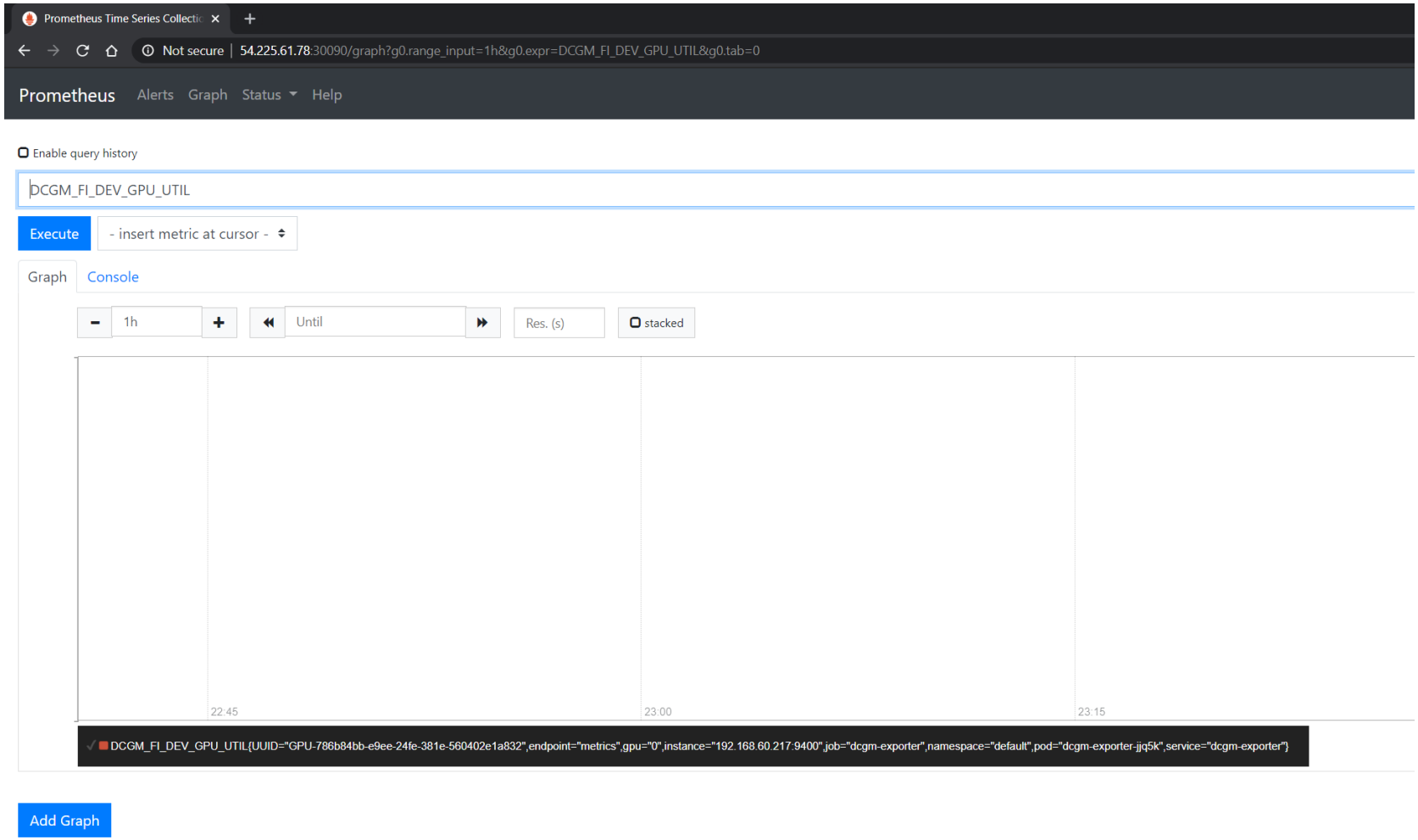
```
$ kubectl get svc -A

NAMESPACE    NAME                                      TYPE      CLUSTER-IP     EXTERNAL-IP
PORT(S)                  AGE
default      dcgm-exporter-2-1603213075                        ClusterIP   10.104.40.255  <none>      9400/
TCP                  7m44s
default      kubernetes                                ClusterIP   10.96.0.1      <none>      443/TCP
49m
kube-system  kube-dns                                  ClusterIP   10.96.0.10     <none>      53/UDP,53/
TCP,9153/TCP       48m
kube-system  kube-prometheus-stack-1603-coredns                  ClusterIP   None        <none>
9153/TCP                 28m
kube-system  kube-prometheus-stack-1603-kube-controller-manager       ClusterIP   None
<none>      10252/TCP                28m
kube-system  kube-prometheus-stack-1603-kube-etcd                 ClusterIP   None        <none>
2379/TCP                 28m
```

```
kube-system   kube-prometheus-stack-1603-kube-proxy              ClusterIP   None        <none>
10249/TCP              28m
kube-system   kube-prometheus-stack-1603-kube-scheduler           ClusterIP   None        <none>
10251/TCP              28m
kube-system   kube-prometheus-stack-1603-kubelet                  ClusterIP   None        <none>
10250/TCP,10255/TCP,4194/TCP   28m
prometheus    alertmanager-operated                               ClusterIP   None        <none>     9093/
TCP,9094/TCP,9094/UDP    28m
prometheus    kube-prometheus-stack-1603-alertmanager             ClusterIP   10.100.20.237
<none>     9093/TCP               28m
prometheus    kube-prometheus-stack-1603-operator                 ClusterIP   10.111.1.27    <none>
8080/TCP,443/TCP          28m
prometheus    kube-prometheus-stack-1603-prometheus               NodePort    10.99.188.46
<none>     9090:30090/TCP            28m
prometheus    kube-prometheus-stack-1603211794-grafana            ClusterIP   10.109.219.60
<none>     80/TCP                 28m
prometheus    kube-prometheus-stack-1603211794-kube-state-metrics      ClusterIP   10.103.250.41
<none>     8080/TCP               28m
prometheus    kube-prometheus-stack-1603211794-prometheus-node-exporter   ClusterIP
10.108.225.36   <none>     9100/TCP               28m
prometheus    prometheus-operated                                 ClusterIP   None        <none>     9090/TCP
28m
```

You can observe that the Prometheus server is available at port 30090 on the node's IP address. Open your browser to `http://<machine-ip-address>:30090`. It may take a few minutes for DCGM to start publishing the metrics to Prometheus. The metrics availability can be verified by typing

`DCGM_FI_DEV_GPU_UTIL` in the event bar to determine if the GPU metrics are visible:




## Using Grafana

You can also launch the Grafana tools for visualizing the GPU metrics.

There are two mechanisms for dealing with the ports on which Grafana is available - the service can be patched or port-forwarding can be used to reach the home page. Either option can be chosen based on preference.

### Patching the Grafana Service

By default, Grafana uses a `ClusterIP` to expose the ports on which the service is accessible. This can be changed to a `NodePort` instead, so the page is accessible from the browser, similar to the Prometheus dashboard.

You can use kubectl patch to update the service API object to expose a `NodePort` instead.

First, modify the spec to change the service type:

```
$ cat << EOF | tee grafana-patch.yaml
spec:
  type: NodePort
  nodePort: 32322
EOF
```

And now use `kubectl patch`:

```
$ kubectl patch svc kube-prometheus-stack-1603211794-grafana \
   -n prometheus \
   --patch "$(cat grafana-patch.yaml)"
```

```
service/kube-prometheus-stack-1603211794-grafana patched
```

You can verify that the service is now exposed at an externally accessible port:

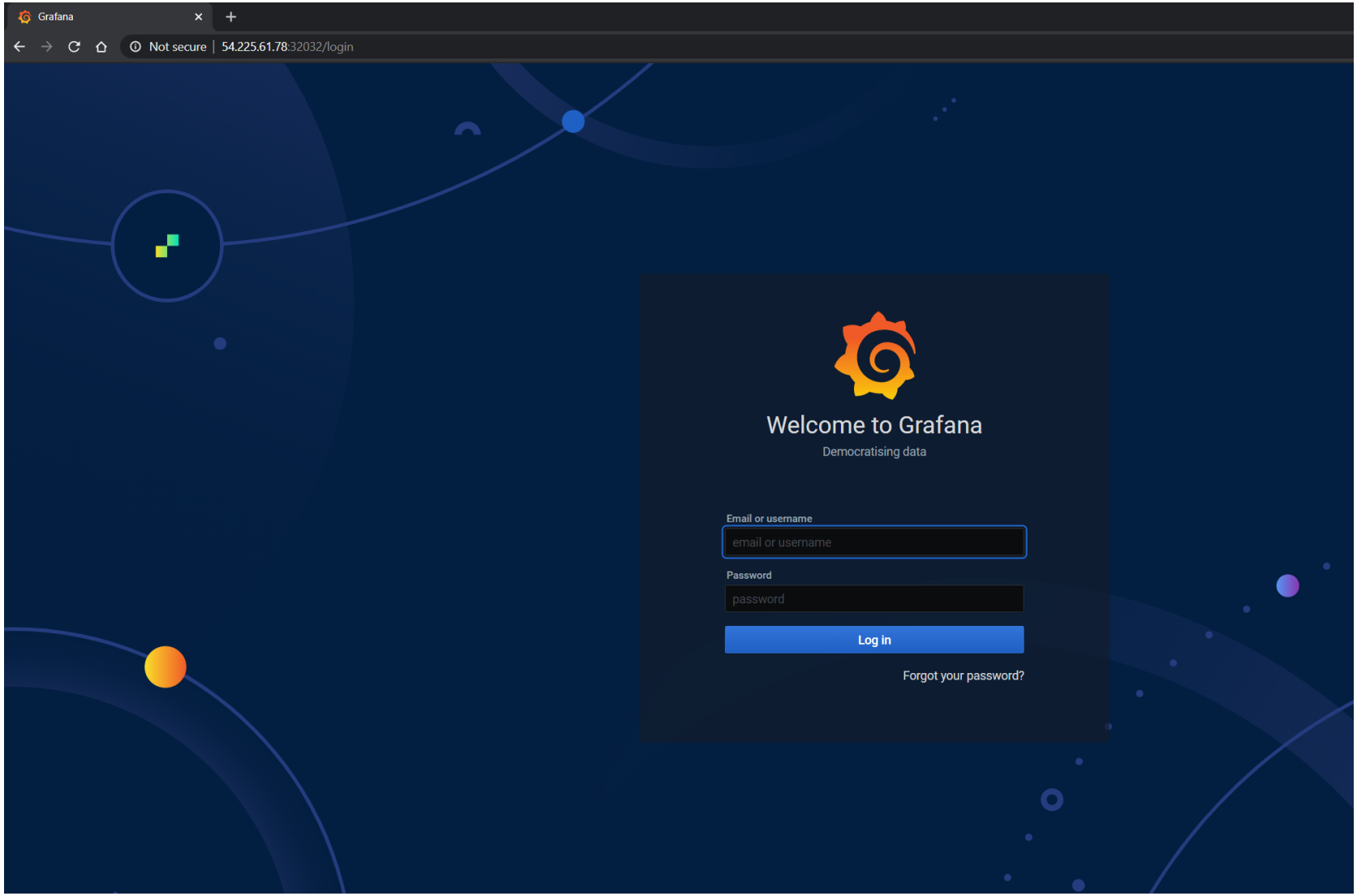
```
$ kubectl get svc -A
```

```
NAMESPACE    NAME                                          TYPE      CLUSTER-IP     EXTERNAL-IP
PORT(S)                AGE
<snip>
prometheus   kube-prometheus-stack-1603211794-grafana              NodePort   10.109.219.60
<none>      80:30759/TCP              32m
```

Open your browser to `http://<machine-ip-address>:30759` and view the Grafana login page. Access Grafana home using the `admin` username. The password credentials for the login are available in the `prometheus.values` file we edited in the earlier section of the doc:

```
## Deploy default dashboards.
##
defaultDashboardsEnabled: true

adminPassword: prom-operator
```




## Port Forwarding

Another method to access the Grafana page would be to use port forwarding.

First, it can be observed that the Grafana service is available at port 80. We will need to port-forward

the service from an abitrary port - in this example, we will forward from port 32322 on our local machine to port 80 on the service (which in turn will forward to port 3000 that the Grafana pod is listening at, as shown below):

```
$ kubectl port-forward svc/kube-prometheus-stack-1603211794-grafana -n prometheus 32322:80
```

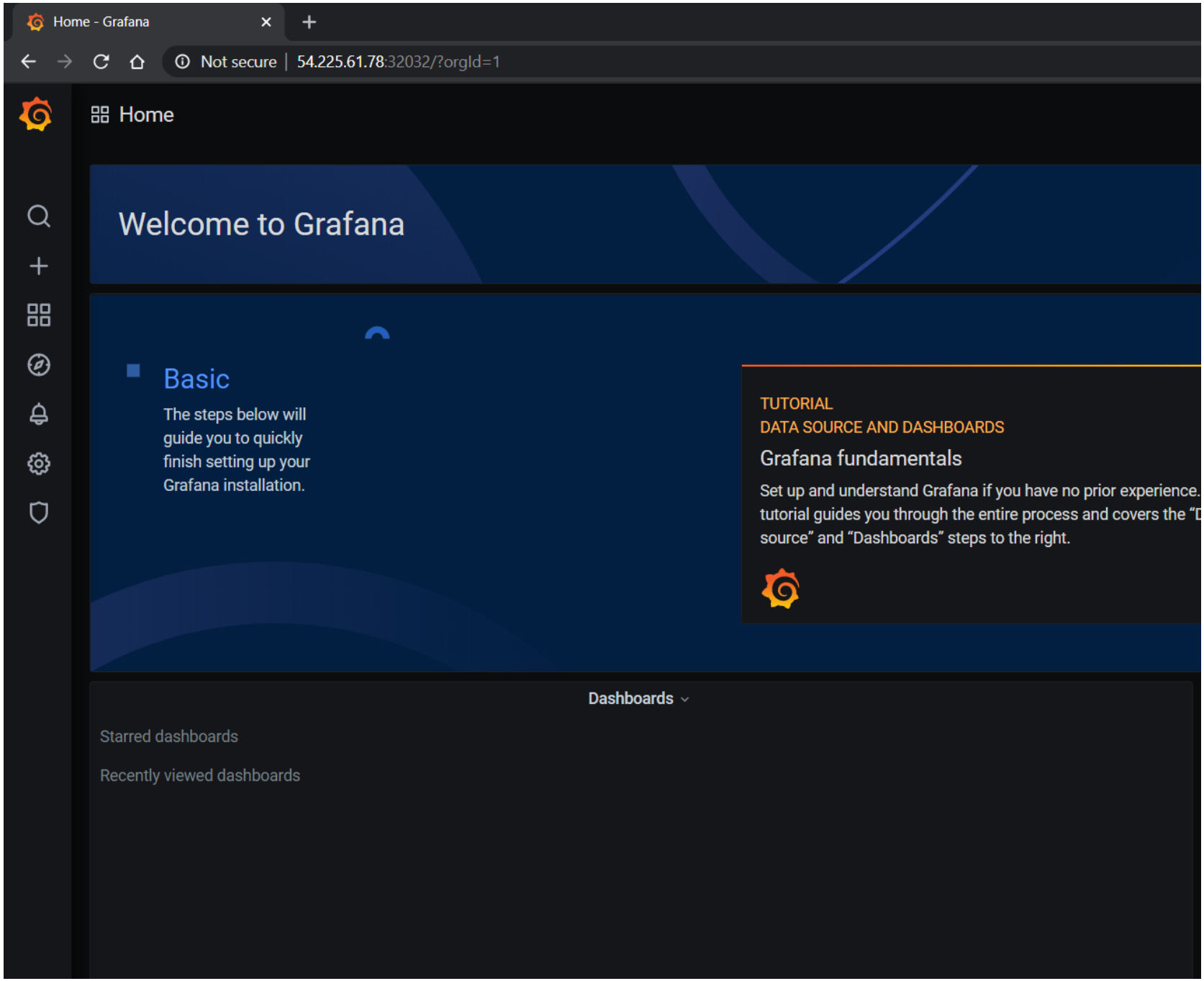
```
Forwarding from 127.0.0.1:32322 -> 3000
Forwarding from [::1]:32322 -> 3000
Handling connection for 32322
```

If your cluster is setup on a cloud instance e.g. AWS EC2, you may have to setup an SSH tunnel between your local workstation and the instance using port forwarding to view the Grafana tool in your local workstation's browser. For example, on Windows you can use PuTTY to open an SSH tunnel and specify the source port as 32322 and destination as `localhost:32322` under the `Tunnels` sub-menu in the SSH menu.

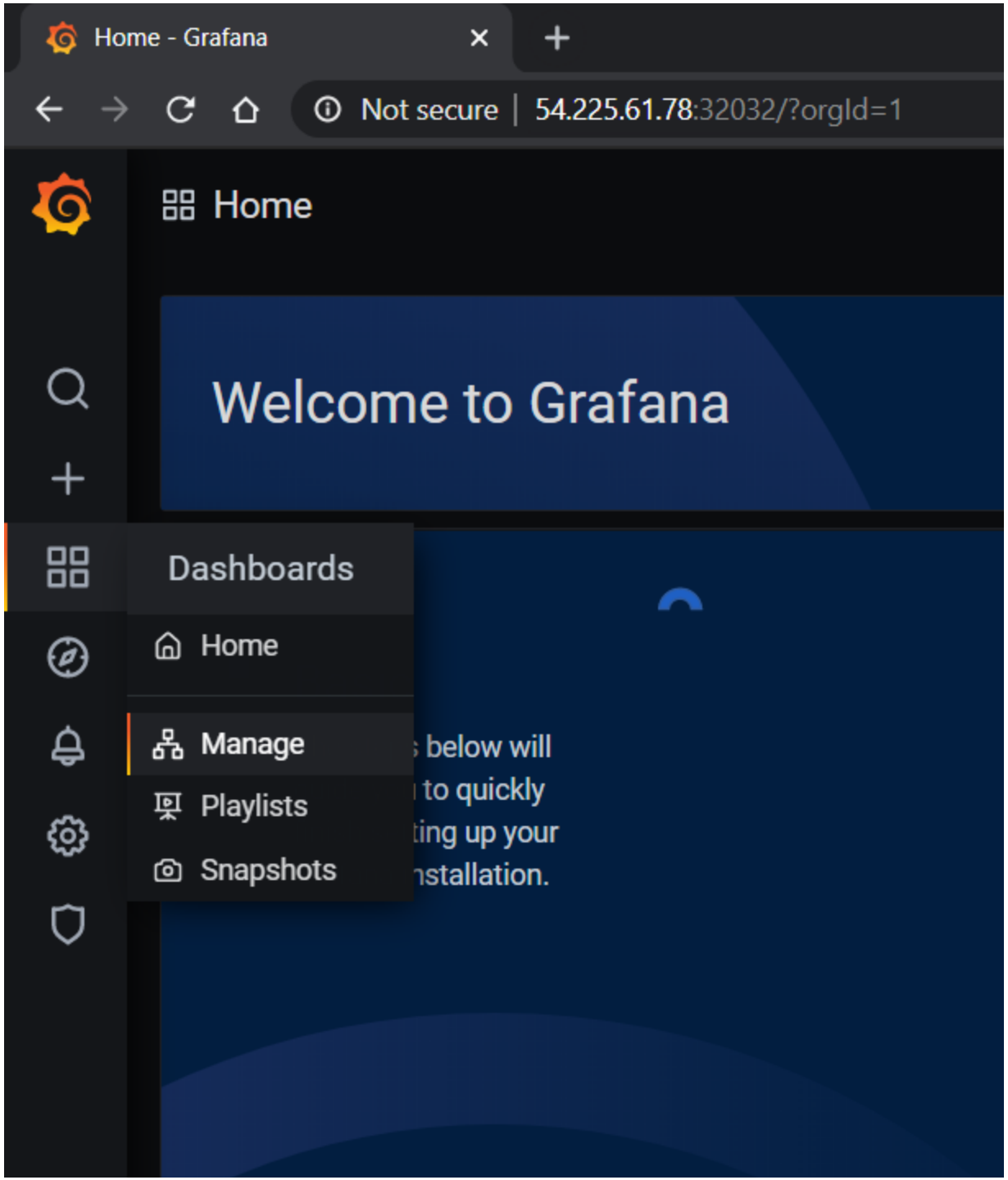
Open your browser and point to `http://localhost:32322/` to view the Grafana login page using the same credentials in the previous section.

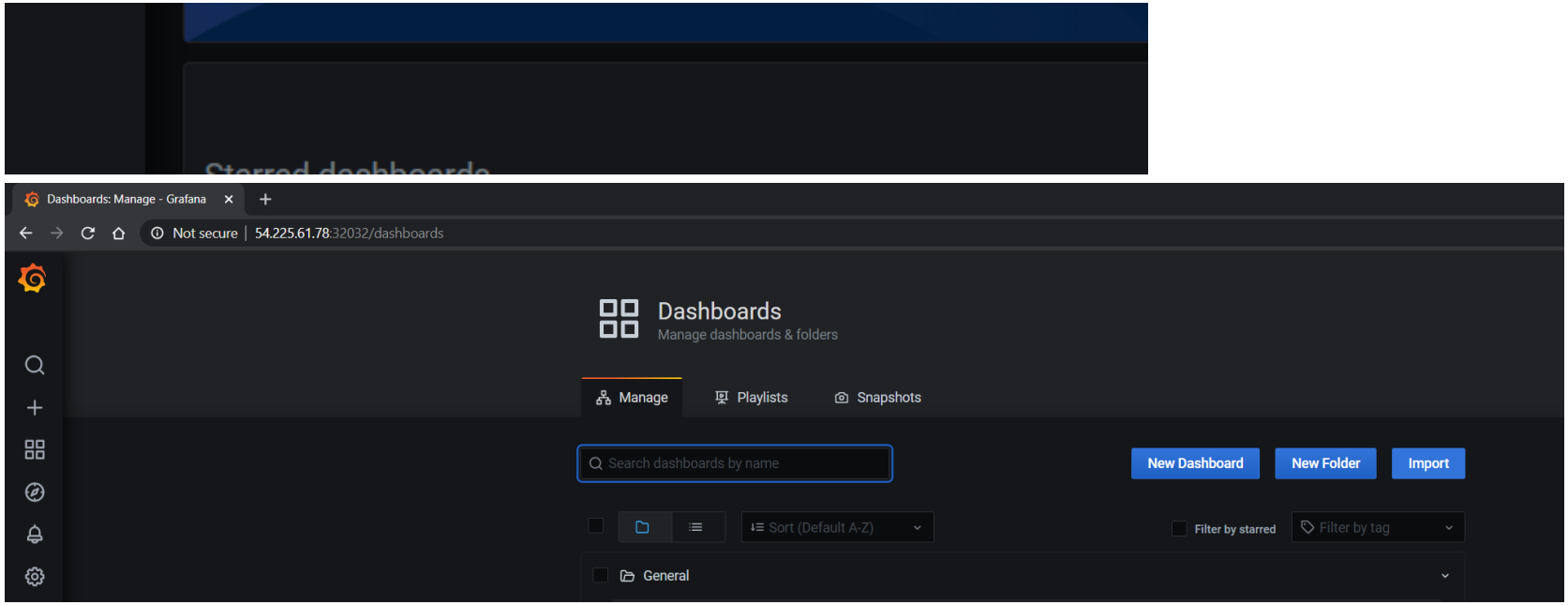
## DCGM Dashboard in Grafana

To add a dashboard for DCGM, you can use a standard dashboard that NVIDIA has made available, which can also be customized.

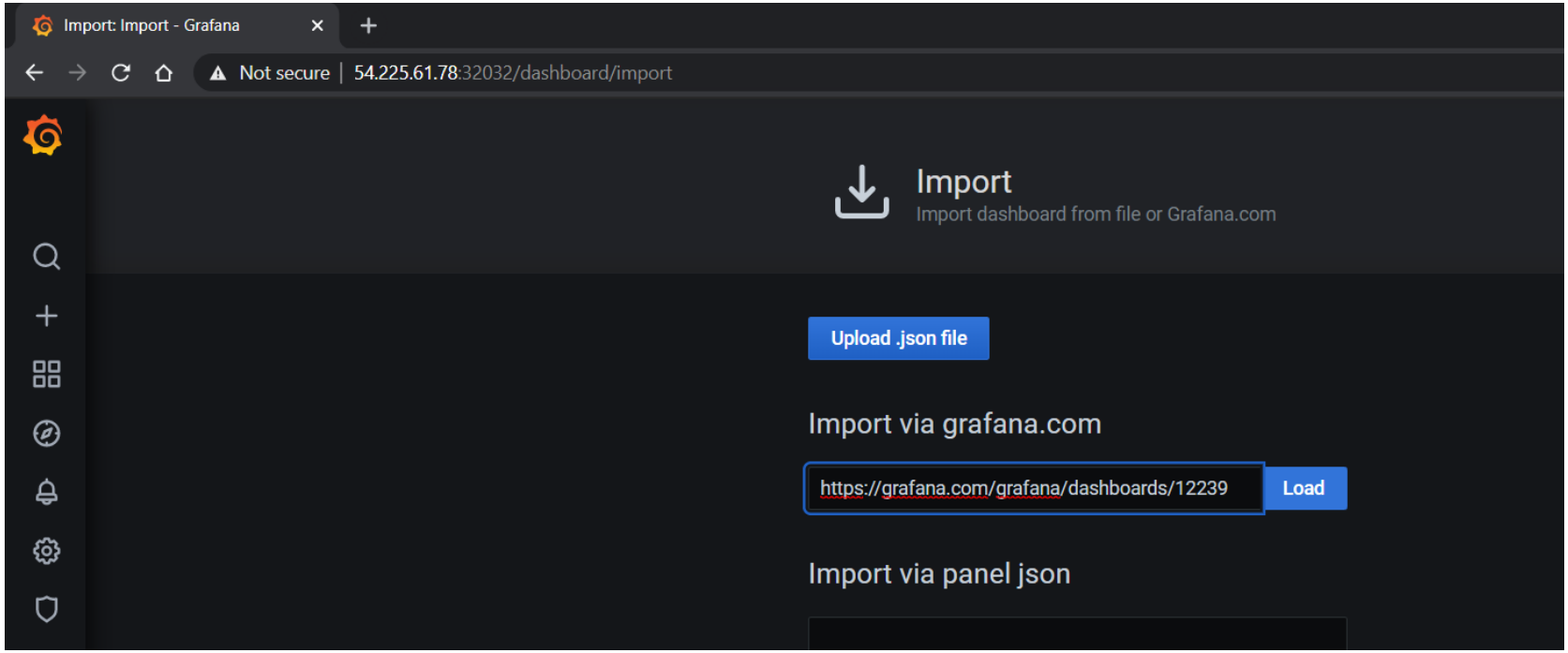


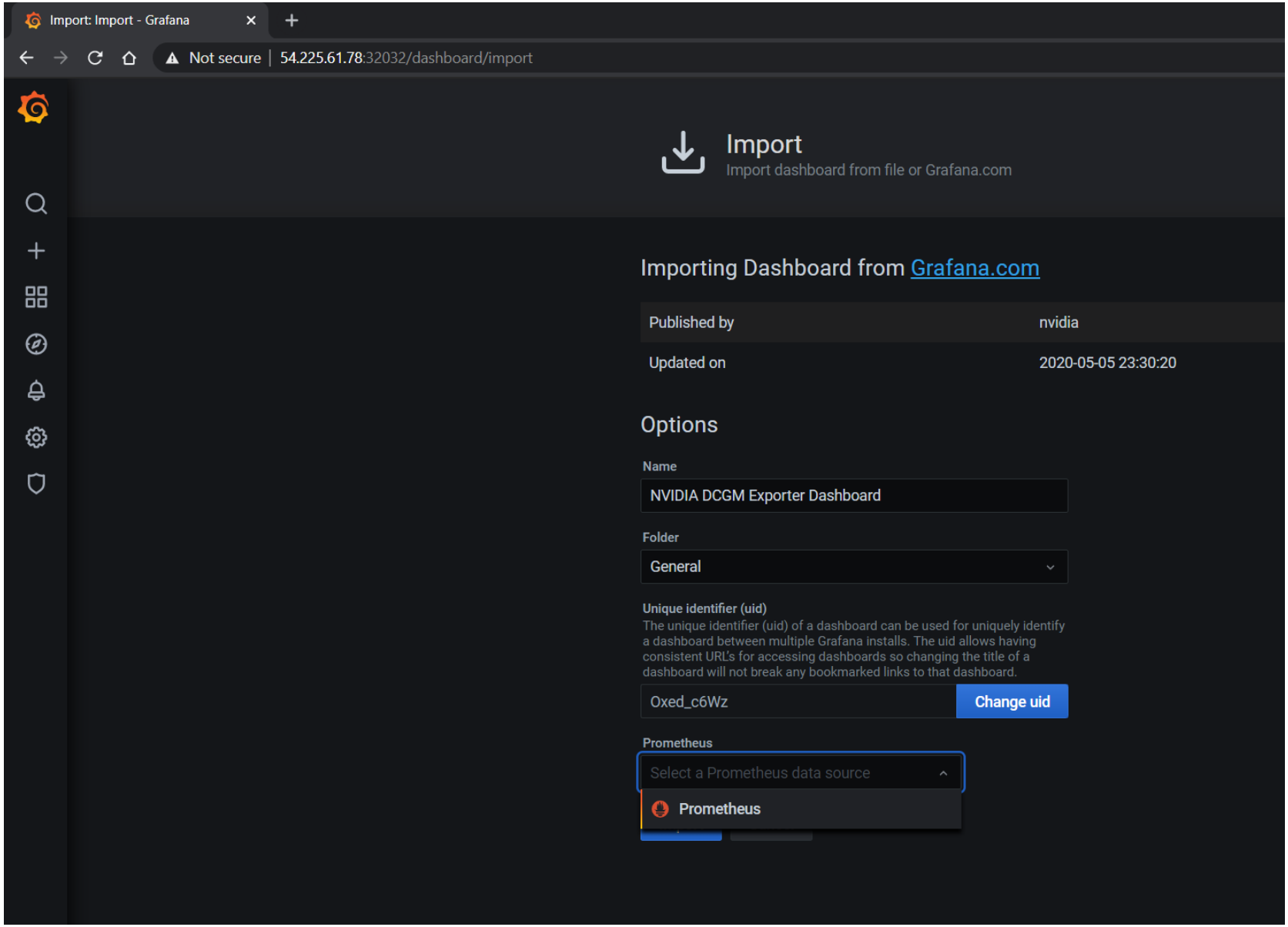

To access the dashboard, navigate from the Grafana home page to Dashboards -> Manage -> Import:


Home - Grafana
Not secure | 54.225.61.78:32032/?orgId=1
Home
Welcome to Grafana
Dashboards
Home
Manage
Playlists
Snapshots
below will
to quickly
ting up your
nstallation.




Import the NVIDIA dashboard from `https://grafana.com/grafana/dashboards/12239` and choose *Prometheus* as the data source in the drop down:


Import: Import - Grafana
Not secure | 54.225.61.78:32032/dashboard/import
Import
Import dashboard from file or Grafana.com
Upload .json file
Import via grafana.com
https://grafana.com/grafana/dashboards/12239
Load
Import via panel json




The GPU dashboard will now be available on Grafana for visualizing metrics:

## Viewing Metrics for Running Applications

In this section, let's run a more complicated application and view the GPU metrics on the NVIDIA dashboard.

We can use the standard *DeepStream Intelligent Video Analytics* Demo available on the NGC registry. For our example, let's use the Helm chart to use the WebUI:

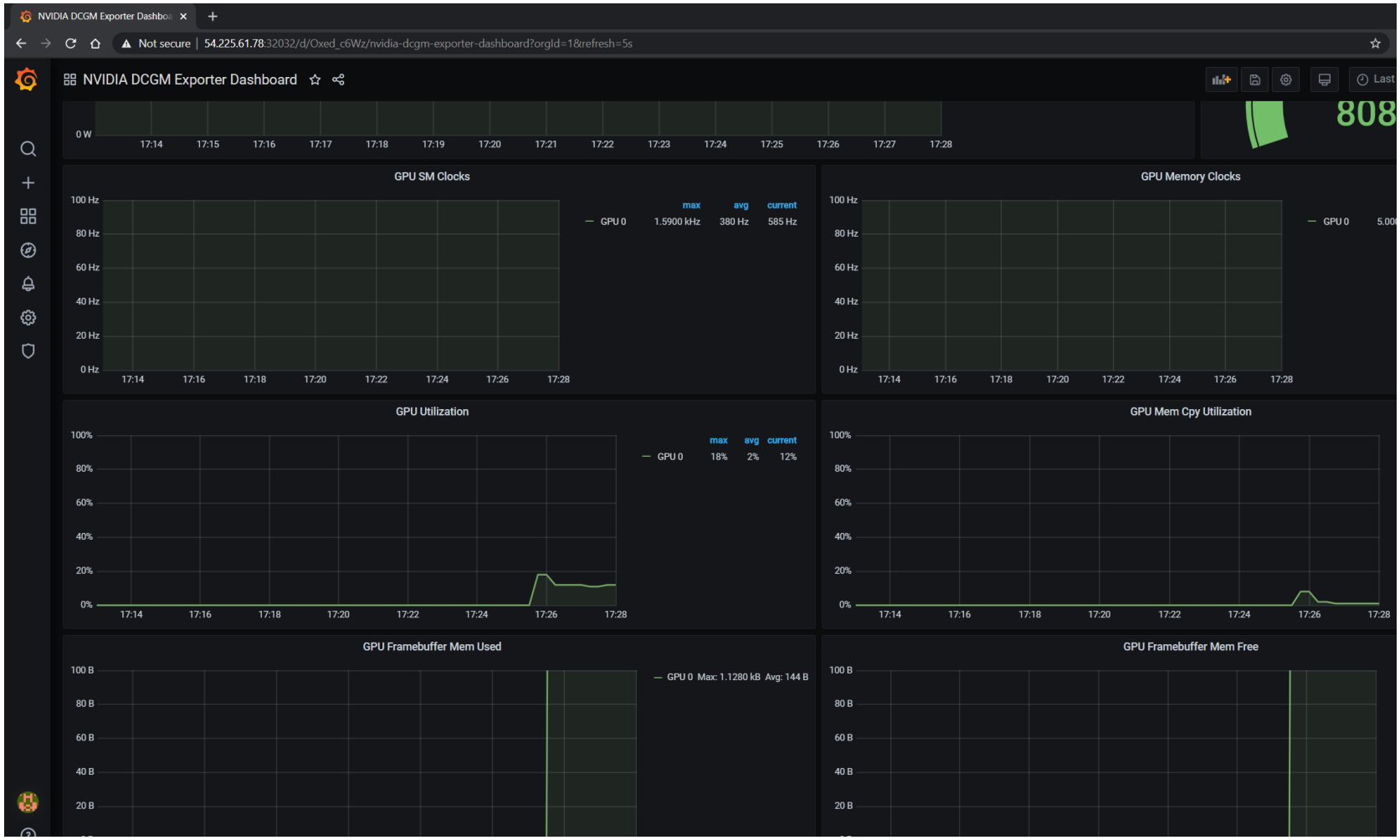
```
$ helm fetch https://helm.ngc.nvidia.com/nvidia/charts/video-analytics-demo-0.1.4.tgz && \
  helm install video-analytics-demo-0.1.4.tgz --generate-name
```

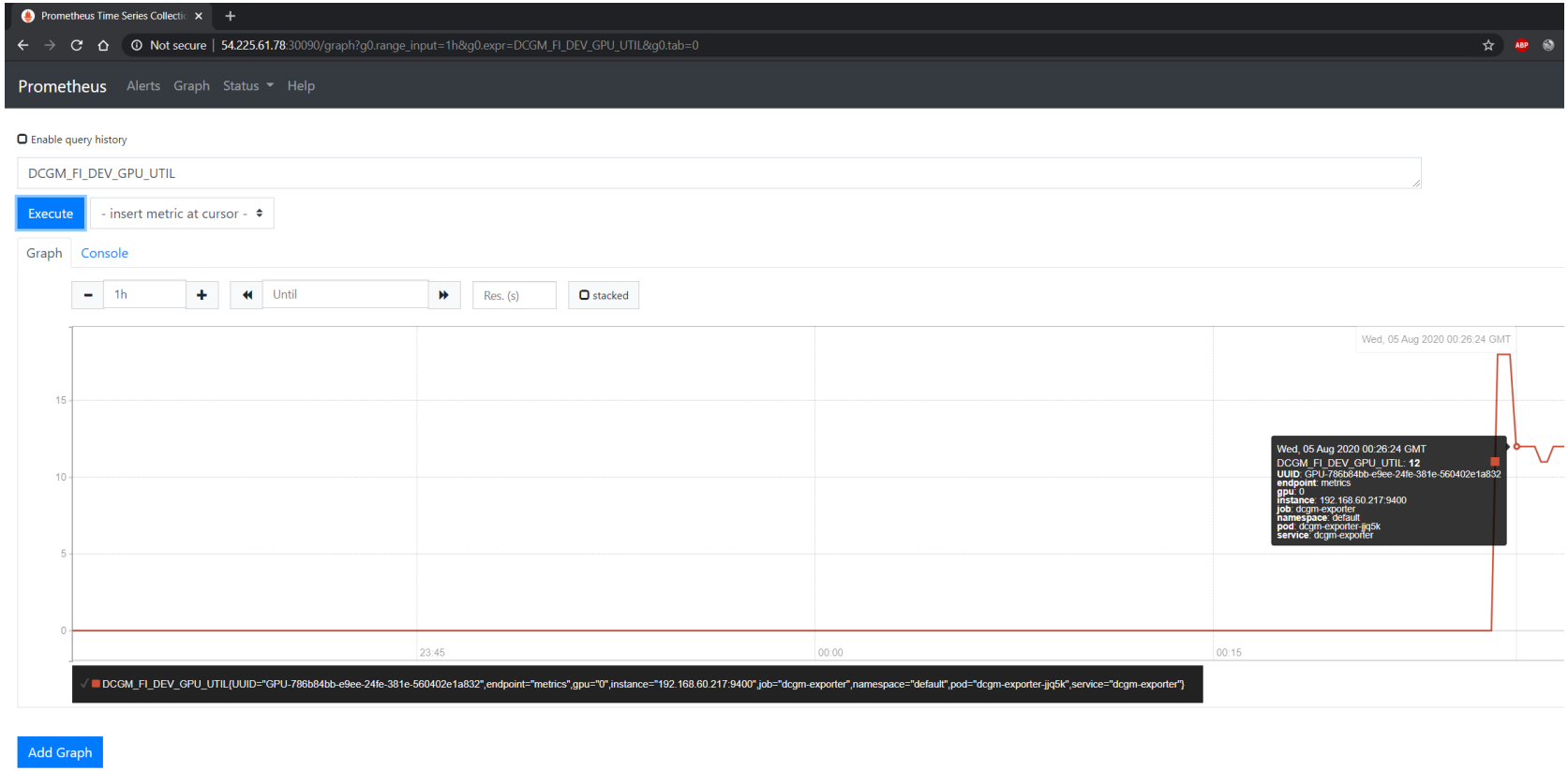
```
NAME: video-analytics-demo-0-1596587131
LAST DEPLOYED: Wed Aug  5 00:25:31 2020
NAMESPACE: default
STATUS: deployed
REVISION: 1
NOTES:
1. Get the RTSP URL by running these commands:
export NODE_PORT=$(kubectl get --namespace default -o jsonpath="{.spec.ports[0].nodePort}"
services video-analytics-demo-0-1596587131)
export NODE_IP=$(kubectl get nodes --namespace default -o
jsonpath="{.items[0].status.addresses[0].address}")
echo rtsp://$NODE_IP:$NODE_PORT/ds-test

2.Get the WebUI URL by running these commands:
export ANT_NODE_PORT=$(kubectl get --namespace default -o
jsonpath="{.spec.ports[0].nodePort}" services video-analytics-demo-0-1596587131-webui)
export NODE_IP=$(kubectl get nodes --namespace default -o
jsonpath="{.items[0].status.addresses[0].address}")
echo http://$NODE_IP:$ANT_NODE_PORT/WebRTCApp/play.html?name=videoanalytics
Disclaimer:
Note: Due to the output from DeepStream being real-time via RTSP, you may experience occasional
hiccups in the video stream depending on network conditions.
```

The demo can be viewed in the browser by pointing to the address following the instructions above.

The GPU metrics are also visible either in the Grafana dashboard or the Prometheus dashboard as can be seen in the following screenshots showing GPU utilization, memory allocated as the application is running on the GPU:




Prometheus Time Series Collecti
Not secure | 54.225.61.78:30090/graph?g0.range_input=1h&g0.expr=DCGM_FI_DEV_GPU_UTIL&g0.tab=0
Prometheus
Alerts
Graph
Status
Help
Enable query history
DCGM_FI_DEV_GPU_UTIL
Execute
- insert metric at cursor -
Graph
Console
1h
Until
Res. (s)
stacked
Wed, 05 Aug 2020 00:26:24 GMT
15
10
5
0
23:45
00:00
00:15
Wed, 05 Aug 2020 00:26:24 GMT
DCGM_FI_DEV_GPU_UTIL: 12
UUID: GPU-786b84bb-e9ee-24fe-381e-560402e1a832
endpoint: metrics
gpu: 0
instance: 192.168.60.217:9400
job: dcgm-exporter
namespace: default
pod: dcgm-exporter-jjq5k
service: dcgm-exporter
DCGM_FI_DEV_GPU_UTIL{UUID="GPU-786b84bb-e9ee-24fe-381e-560402e1a832",endpoint="metrics",gpu="0",instance="192.168.60.217:9400",job="dcgm-exporter",namespace="default",pod="dcgm-exporter-jjq5k",service="dcgm-exporter"}
Add Graph